Abhishek M

Data Scientist

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SUMMARY

Data Scientist with around 7 years of experience in Healthcare, e-commerce, Automobile and Insurance domain. Skilled at performing **Data Extraction, Data Screening, Data Cleaning, Data Exploration**, **Data Visualization** and **Statistical Modelling** of varied datasets, structured and unstructured, as well as implementing large-scale **Machine Learning** and **Deep Learning Algorithms** to deliver resourceful insights and inferences significantly impacting business revenues and user experience.

- Experienced in facilitating the entire lifecycle of a data science project: **Data Extraction, Data Pre-Processing**, **Feature Engineering, Algorithm Implementation & Selection, Back Testing** and **Validation**.
- Expert at working with **Statistical Tests Parametric tests**: t-test, ANOVA along with **Non-parametric tests**: Chi-squared tests, Mann-Whitney U & Kruskal-Wallis test.
- Skilled in using Python libraries **NumPy, Pandas** for performing **Exploratory Data Analysis**.
- Proficient in Data transformations using **log, square-root, reciprocal, cube root, square and complete box-cox transformation** depending upon the dataset.
- Adept at handling **Missing Data** by exploring the causes like MAR, MCAR, MNAR and analyzing **Correlations** and similarities, introducing dummy variables and various **Imputation** methods.
- Experienced in Machine Learning techniques such as **Regression and Classification** models like **Linear Regression, Logistic Regression,** Decision Trees, Support Vector Machine using **scikit-learn** on Python.
- In-depth Knowledge of **Dimensionality Reduction** (PCA, LDA)**, Hyper-parameter tuning, Model Regularization** (Ridge, Lasso, Elastic net) and **Grid Search techniques** to optimize model performance.
- Skilled at Python, SQL, R and **Object Oriented Programming (OOP)** concepts such as Inheritance, Polymorphism, Abstraction, Encapsulation**.**
- Working knowledge of Database Creation and maintenance of Physical data models with **Oracle, DB2 and SQL server** databases as well as normalizing databases up to third form using **SQL** functions.
- Experience in Web Data Mining with Python's **ScraPy** and **BeautifulSoup** packages along with working knowledge of **Natural Language Processing (NLP)** to analyze text patterns.
- Proficient in **Natural Language Processing (NLP)** concepts like Tokenization, Stemming, Lemmatization, Stop Words, Phrase Matching and libraries like **SpaCy** and **NLTK**.
- Skilled in **Big Data** Technologies like **Apache Spark** (PySpark, Spark Streaming, MLlib), **Hadoop** Ecosystem (MapReduce, HDFS, HIVE, Kafka, Ambari)
- Proficient in **Ensemble Learning** using Bagging, Boosting (AdaBoost, xGBoost) & Random Forests; clustering like **K-means**.
- Experienced in developing **Supervised Deep Learning** algorithms which include Artificial Neural Networks, Convolution Neural Networks, Recurrent Neural Networks, LSTM, GRU and **Unsupervised Deep Learning** Techniques like Self-Organizing Maps (SOM's) in **Keras** and **TensorFlow**.
- Built and deployed recurrent neural network architecture called **LSTM** in one of the projects to improve the accuracy of the model, also have knowledge of Deep Learning approaches such as traditional **Artificial Neural Network** and **Convolutional Neural Network**.

- Skilled at **Data Visualization** with **Tableau, PowerBI**, Seaborn, Matplotlib, ggplot2, Bokeh and interactive graphs using **Plotly & Cufflinks.**
- Knowledge of **Cloud services** like Amazon Web Services **(AWS)** and **Microsoft Azure** for building, training and deploying scalable models.
- Proficient in using **PostgreSQL**, **Microsoft SQL server** and **MySQL** to extract data using multiple types of **SQL Queries** including **Create**, **Join, Select, Conditionals, Drop, Case** etc.

SKILLS

Languages and Platforms	Python (Numpy, Pandas, Scikit-learn, Tensorflow, etc.), Spyder, JuPyter lab, R Studio(ggplot2, dplyr, lattice, highcharter etc), SQL, SAS.
Analytical Techniques	**Regression Methods:** Linear, Polynomial, Decision Trees; **Classification:** Logistic Regression, K-NN, Naïve Bayes, Support Vector Machines (SVM); **Ensemble Learning:** Random Forests, Gradient Boosting, Bagging; **Clustering:** K-means clustering, Hierarchical clustering; **Deep Learning:** Artificial Neural Networks, Convolutional Neural Networks, Recurrent Neural Networks; **Dimensionality Reduction:** Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA); **Time-series Forecasting:** Using Python Statsmodels and **R** for time-series modelling.
Database	SQL, PostgreSQL, MongoDB, Microsoft SQL Server, NoSQL, Oracle
Statistical Tests	Hypothesis Testing, ANOVA, z-test, t-test, Chi-Squared Fit test
Validation Techniques	Monte Carlo simulations, k-fold cross validation, A/B Testing
Optimization Techniques	Gradient Descent, Stochastic Gradient Descent, Gradient Optimization – Momentum, RMSProp, Adam
Big Data	Apache Hadoop, HDFS, MapReduce, Apache Spark, HiveQL, Pig, Kafka
Data Visualization	Tableau, Microsoft PowerBI, ggplot2, Matplotlib, Seaborn, Bokeh, Plotly
Data Modeling	Entity Relationship Diagrams (ERD), Snowflake Schema, SPSS Modeler
Operating Systems	Microsoft Windows, iOS, Linux Ubuntu
Database Systems	SQL Server, Oracle, MySQL, **Teradata** Processing System, **NoSQL** (**MongoDB**, **HBase, Cassandra**), **AWS** (**DynamoDB**, **ElastiCache**)

PROJECTS

eBay INC. **Location: Salt lake City, Utah**
Data Scientist
(May2019-Present)

Description:
Goal was to derive factors affecting opening new stores and suggesting insights which can help business owners considering opening stores and company increasing their revenue. This was achieved through analyzing the spending habits and spending capacity of customers, by classifying them into clusters using **K-NN** and K means clustering and then using various predictive regression models to forecast potential spending habits of customers.

Responsibilities:
- Worked closely with Business Analyst to understand business requirements and start deriving solutions accordingly.

- Worked with Data Engineers and Data Analysts into a cross functional team for the deployment of models and working of the projects.
- Extracted data from HTML and XML files by web-scraping through customer reviews using **Beautiful Soup**, also pre-preprocessed raw data from the company's data warehouse.
- Performed **Data collection, Data cleaning, Feature scaling, Feature engineering, Validation, Visualization, Data Resampling**, report findings, develop strategic uses of data by **Python** libraries like **NumPy, Pandas, Scipy, Scikit-Learn, TensorFlow.**
- Involved in various pre-processing phases of text-data like **Tokenization, Stemming, Lemmatization** and converting the raw text data to structured data.
- Performed **collaborative filtering** to generate item recommendations. Rank-based and content-based recommendations were used to address the problem of cold start.
- Performed data post processing using **NLP** techniques like **TF-IDF**, Word2Vec & BOW to identify the most pertinent Product Subject Headings terms that describe items.
- Performed **Data Visualization** using RStudio, used **ggplot2, lattice, highcharter, Leaflet, Plotly & Cufflinks, sunburstR, RGL** to make interesting plots.
- Implemented various statistical techniques to manipulate the data like missing **data imputation**, **Principal Component Analysis, tSNE** for dimension-reduction.
- Performed **Naïve Bayes, K-NN, Logistic Regression, Random Forest, SVM** and **KMeans** to categorize customers into certain groups.
- Performed **Linear Regresion** onto the classified clusters of customers that were deduced from clustering through **K-NN** and K-means clustering.
- Employed statistical methodologies such as A/B test, experiment design and hypothesis testing.
- Used AWS **Sagemaker** to train model using protobuf and deploy the model owing to its relative simplicity and computational efficiency over Beanstalk.
- Employed various metrics such as **Cross-Validation, Confusion Matrix, ROC and AUC** to evaluate the performance of each model.

Environment: Python 3.6 (NumPy, Pandas, Matplotlib), PySpark 2.4.1, AWS, SQL Server, RStudio

Anthem INC. **Location: Norfolk, VA**
Data Scientist
(March 2018-May2019)

Description
Anthem is a health insurance service provider which specializes in health and wellness for its users. My project at Anthem involved predicting various levels of risk factors using machine learning techniques and Hypothesis Testing for the company approving medical insurance of the applicants based on their medical, insurance and employment history.

Responsibilities:
- Performed data collection, data cleaning, data profiling, data visualization and report creating.
- Extracted required medical data from **Azure Data Lake Storage** into **PySpark** dataframe for further exploration and visualization of the data to find insights and build prediction model.
- Performed data cleaning on the medical dataset which had missing data and extreme outliers from **PySpark** data frames and explored data to draw relationships and correlations between variables.

- Implemented data pre-processing using **Scikit-Learn**. Steps include Imputation for missing values, Scaling and logarithmic transform, one hot encoding etc.
- Analyzed the applicant's medical data to find various relations which were further plotted using Alteryx and **Tableau** to get better understanding on the available data.
- Utilized Python's data visualization libraries like **Matplotlib** and **Seaborn** to communicate findings to the data science, marketing and engineering teams.
- Performed **univariate, bivariate and multivariate analysis** on the BMI, age and employment to check how the features were related in conjunction to each other and the risk factor.
- Trained several machine learning models like **Logistic Regression, Random Forest** and **Support vector machines (SVM)** on selected features to predict Customer churn.
- Improved model accuracy by 5% by introducing Ensemble techniques: **Bagging, Gradient, Xtreme Gradient** and **Adaptive Boosting**.
- Worked on Statistical methods like data driven **Hypothesis Testing** and **A/B Testing** to draw inferences, determined significance level and derived P-value, and to evaluate the impact of various risk factors.
- Furthered Hypothesis testing by evaluating Errors (Type 1 and Type 2) to eliminate skewed inferences.
- Implemented and tested the model on **AWS EC2** and collaborated with development team to get the best algorithms and parameters.
- Prepared data-visualization designed dashboards with Tableau, and generated complex reports including summaries and graphs to interpret the findings to the team.

Environment: Python (NumPy, Pandas, Matplotlib, Sk-learn), AWS, Jupyter Notebook, HDFS, Hadoop MapReduce, PySpark, Tableau, SQL

OLA Cabs **Location: Mumbai, India**
Junior Data Scientist
(November 2015-December 2017)

Description:
The data contained offer history and purchase behaviour of customers. The dataset was used to figure out how customers respond to different types of offers and how much they spent with and without those offers with aim of predicting what their response would be to offer types as well as their Life Time Value (CLV).

Responsibilities:
- Performed exploratory data analysis and data cleaning along with data visualization using Seaborn and Matplotlib.
- Performed feature engineering to create new features and determine transactions associated with completed offer.
- Performed data visualization using Matplotlib and Seaborn from data using features like age, income, membership duration, etc.
- Built machine learning models using customer models and transaction data.
- Built a classification model to classify customers for promotional deals to increase likelihood of purchase using **Logistic Regression** and **Decision Tree Classifier**.
- Tested out performance of classifiers **like Logistic Regression, Naïve Bayes, Decision tress** and **Support vector classifiers**.
- Employed ensemble learning techniques like Random forests and Ada Gradient Boosting to improve the model by 15%.
- Picked the final model using **ROC & AUC** and fine-tuned the hyper parameters of the above models using Grid Search to find the optimum values.
- Using k-fold cross validation to test and verify the model accuracy.
- Prepared dashboard in PowerBI to summarize the model and show the summary of model's measure.

Environment: Python (NumPy, Pandas, Matplotlib, SkLearn), Tidyverse, R, MySQL, PgAdmin

Big Basket **Location: Ahmedabad, India**

Data Analyst
(July 2013-October 2015)

Description:
Big Basket is India's largest online food and grocery store. My responsibilities included drawing statistical inferences and applying optimization techniques related to Economic Order Quantity. Analysed the model to optimize inventory and fill rate while reducing re-order cost and time. Also responsible for maintaining databases related to inventory.

Responsibilities:
- Drew statistical inferences using **t-tests, ANOVA, chi-sq tests** and performed Post-Hoc Analysis using Tukey's HSD and Bonferroni correction to assess difference across levels of raw material categories, test significance of proportional differences and assess whether sample size is large enough to detect the differences.
- Provided statistical insights into **semi-deviation** & **skewness-to-kurtosis** ratio to guide vendor decisions and inferences into optimum pricing for order quantities.
- Performed **Data Analysis** on target data after transfer to Data Warehouse.
- Developed interactive executive dashboards using **PowerBI** and **Excel VBA** to provide a reporting tool that facilitates organizational metrics and data
- Worked with Data Architects on **Dimensional Model** with both Star and Snowflake Schemas utilized
- Created ETL solution using **Informatica** tool to read Product & Order data from files on shared network into **SQL Server** database.
- Created Database designs through **data-mapping** using ER diagrams and normalization upto the 3rd normal form and extracted relevant data whenever required using joins in Postgre SQL and Microsoft SQL Server.
- Conducted **data preparation** and outlier detection using Python.
- Worked with Team manager to develop a lucrative system of classifying auditions and vendor's best fitting for the company in the long run.
- Presented executive dashboards and scorecards to visualize and present trends in the data using **Excel** and **VBA-Macros**.

Environment: Microsoft Office, Microsoft PowerBI, Oracle SQL, Informatica, SPSS.

Education

Bachelor of Technology in Electrical Engineering.
Pandit Deendayal Petroleum University, India.

Master of Science in Technology Management with Concentration area of IT and Big Data.
University of Bridgeport, Bridgeport, Connecticut.